1. Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations (Point 5 in the list)

This is the most core competency for this question. Students need to **recognize** a key geometric pattern: the area of an L-shaped or irregular polygon can be calculated by enclosing it within a complete rectangle, and the total area of this outer rectangle is equal to the sum of the areas of all the internal rooms (including the terrace and walls).

- **Specific Analysis:** The traditional area calculation method (decomposing the irregular shape into multiple small rectangles and adding them up) requires measuring up to a dozen lengths. This question asks for a "more efficient method," which requires students to recognize that although the apartment layout is irregular, the entire structure can be treated as one large rectangle. By measuring the total external length and total external width of this large rectangle, the total area can be calculated in one step.

2. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation (Point 4 in the list)

Students must understand the floor plan layout, the scale (1 cm represents 1 m), and the definition of "total floor area" (including the terrace and walls) to correctly model the problem.

- **Specific Analysis:** Understanding the scale is crucial for converting measurements on the drawing into actual meters. Understanding that wall areas must be included means measurements must be taken to the outermost edges of the exterior walls, not just internal room dimensions.

3. Select a mathematical description or a representation that describes a problem (Point 1 in the list) / Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem (Point 10 in the list)

This question requires students to mark the four necessary lengths on the floor plan.

- **Specific Analysis:** These four lengths represent the total length and total width of the enclosing rectangle. Students need to visualize the abstract mathematical concept (rectangle area formula $Area = Length \times Width$, using variables L and W) onto the physical image, marking L and W on the plan to represent their chosen mathematical model.

4. Identify the constraints, assumptions simplifications in a mathematical model (Point 12 in the list)

This "enclosing rectangle" method is a simplified model. Students need to understand the assumptions of this model.

- **Specific Analysis:** This method assumes that the overall shape of the apartment is a perfect rectangle and that all corners are right angles. Students need to understand the slight estimation error this simplified geometric model might introduce and how it is more efficient than other more precise but cumbersome methods.

1. 识别问题情境的数学结构（包括规律、关系和模式）（列表中的第 5 点）

这是此题最核心的能力。学生需要**识别**到一个关键的几何学规律：一个 L 形或不规则多边形的面积可以通过将其外接成一个完整的矩形来计算，并且该外接矩形的总面积等于所有内部房间（包括露台和墙壁）的总面积。

- **具体分析：** 传统的面积计算方法（将不规则图形分解成多个小矩形并相加）需要测量多达十几个长度。该题目要求找到一种“更有效的方法”，这需要学生识别出，公寓的布局虽然不规则，但整体可以被视为一个大的矩形，通过测量这个大矩形的外部总长度和总宽度，就可以一步计算出总面积。

2. 阅读、解码和理解陈述、问题、任务、对象或图像以创建情境模型 (列表中的第 4 点)

学生必须理解平面图的布局、比例尺（1 cm 代表 1 m）、以及“总建筑面积”的定义（包括露台和墙壁），才能正确构建问题模型 (p. 1)。

- **具体分析：** 理解比例尺对于将图纸上的测量值转换为实际米数至关重要。理解需要包含墙壁面积意味着必须测量到建筑外墙的最外侧边缘，而不是内部房间的尺寸。

3. 选择描述问题的数学描述或表示形式 (列表中的第 1 点) / 使用适当的变量、符号、图表等数学工具来描述数学结构和/或关系 (列表中的第 10 点)

此题要求学生在平面图上标记出所需的四个长度。

- **具体分析：** 这四个长度代表了外接矩形的总长度和总宽度。学生需要将抽象的数学概念（矩形面积公式 $Area = Length \times Width$，使用变量 L 和 W）具体化到物理图像上，通过在图上标记 L 和 W 来表示他们选择的数学模型。

4. 识别模型中的限制、假设和简化 (列表中的第 12 点)

这种“外接矩形”的方法是一种简化模型。学生需要理解这个模型的假设。

- **具体分析：** 这种方法假设公寓的整体形状是一个完美的矩形，并且所有墙角都是直角。学生需要在使用这个简化的几何模型时，理解它可能带来的轻微估算误差，以及它如何比其他更精确但更繁琐的方法更有效。